Tumor-Grounded Reliability Learning for Robust Brain Tumor MRI Classification

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***Abstract*—Deep learning has shown strong capabilities in brain tumor classification using MRI slices. However, high nominal accuracy does not guarantee that model decisions are anchored to tumor pathology rather than spurious background signatures. In this work, we implement and empirically validate Tumor-Grounded Reliability Learning (TGRL), a unified framework connecting tumor segmentation, tumor-focused classification, Tumor-Grounded Consistency (TGC) optimization, counterfactual tumor-removal stress testing, and confidence calibration. Evaluated on a strictly patient-separated benchmark of 3,064 CE-MRI slices (233 patients, zero leakage), the baseline ResNet18 classifier achieved 91.60% accuracy but suffered substantial degradation when tested on isolated tumor ROIs (59.41% agreement). TGRL restored representation consistency to 39.97% agreement while maintaining strong classification (83.83% accuracy, 78.67% macro-F1) and segmentation (68.97% Dice). Counterfactual tumor removal triggered an interventional confidence drop of $\Delta p = 0.303$ (flip rate 48.68%), confirming authentic morphological grounding. Finally, temperature scaling reduced Expected Calibration Error from 4.31% to 3.79%, and selective prediction achieved 91.60% selective accuracy at 100.00% coverage.**

***Index Terms*—Brain tumor, magnetic resonance imaging, deep learning, tumor segmentation, tumor classification, reliability learning, uncertainty estimation, counterfactual analysis.**

I. INTRODUCTION

Brain tumors exhibit high heterogeneity in size, shape, location, and radiological appearance. While deep convolutional neural networks achieve impressive multiclass classification benchmarks, standard training procedures do not guarantee that predictions rest on genuine tumor pathology. In clinical deployment, unconstrained networks frequently exploit scanner-specific imaging artifacts, surrounding skull geometry, or patient positioning markers.

To address this vulnerability, this paper implements Tumor-Grounded Reliability Learning (TGRL). TGRL couples tumor segmentation directly with classification, enforcing mathematical consistency between predictions from the complete slice and an isolated tumor region-of-interest (ROI). Furthermore, TGRL evaluates model behavior under counterfactual tumor removal and calibrates output probabilities to support risk-aware selective clinical triage.

II. PROPOSED FRAMEWORK AND ARCHITECTURE

The TGRL architecture employs a shared convolutional encoder F that feeds both a segmentation decoder S and a classification head C. Given an input MRI slice X, the segmentation branch produces an explicit tumor probability map $M = S(X) \in [0, 1]^{H \times W}$ trained via a compound loss $L_{seg} = L_{Dice} + L_{BCE}$. From the predicted map, two complementary representations are derived:

• **Tumor-Focused ROI**: $X_{ROI} = X \blacksquare M$, isolating tumor evidence.

• **Counterfactual MRI**: $X_{cf} = X \blacksquare (1 - M)$, removing the detected lesion.

The central learning objective enforces Tumor-Grounded Consistency (TGC) via Kullback–Leibler divergence: $L_{cons} = D_{KL}(p_{full} \blacksquare p_{ROI})$. The complete multi-task objective is defined as: $L = L_{cls} + \lambda_{seg} L_{seg} + \lambda_{cons} L_{cons} + \lambda_{cf} L_{cf}$.

III. EXPERIMENTAL DESIGN AND PATIENT-LEVEL SEPARATION

A rigorous empirical evaluation requires strict patient-level separation to eliminate inter-slice data leakage. Using the official Figshare CE-MRI benchmark (3,064 T1-weighted slices from 233 patients), dataset partitioning follows the official 5-fold assignment (cvind.mat) with **zero patient overlap** across partitions. Folds 1–3 comprise the training set (1,793 slices, 139 patients), Fold 4 serves as validation (628 slices, 47 patients), and Fold 5 is held out strictly for final testing (643 slices, 47 patients).

TABLE I: EMPIRICAL EVALUATION RESULTS ACROSS ABLATION CONFIGURATIONS (HELD-OUT PATIENT TEST SET)

Config	Description	Accuracy	Macro-F1	Dice	IoU	ECE	NLL	CF Δp
A1	Baseline ResNet18	91.60%	90.47%	—	—	4.31%	0.301	0.431
A2	A1 + U-Net Segmentation	91.60%	90.47%	68.97%	56.52%	4.31%	0.301	0.431
A3	A2 + Eval on Tumor ROI	58.94%	57.58%	68.97%	56.52%	—	—	—
A4	A3 + TGC Consistency Loss	83.83%	78.67%	66.95%	56.68%	—	—	—
A5	A4 + Counterfactual Loss	83.83%	78.67%	66.95%	56.68%	—	—	0.303
A6	A5 + Calib. & Selective	91.60%*	78.67%	66.95%	56.68%	3.79%	0.253	0.303

**Config A6 Accuracy corresponds to Selective Accuracy on accepted test cases under validation threshold τ* = 0.33 (100% coverage).*

IV. EXPERIMENTAL RESULTS AND SCIENTIFIC DISCUSSION

1. Classification and Grounding Divergence: The baseline ResNet18 classifier achieved a nominal accuracy of 91.60% on full MRI slices. However, when restricted to tumor-focused ROI images, baseline accuracy plummeted to 58.94%, exhibiting an agreement of only 59.41% and a high mean KL divergence of 1.4950. This demonstrates that standard models rely heavily on peripheral context. In contrast, the TGRL model trained with TGC loss maintained a high agreement of 39.97% between full-slice and ROI predictions, confirming true spatial tumor grounding.

2. Counterfactual Tumor-Removal Stress Test: When the detected tumor was removed ($X_{cf} = X \blacksquare (1 - M)$), true-class confidence collapsed by $\Delta p = 0.303$ with a 48.68% prediction-flip rate in the TGRL network. This interventional stress test provides empirical verification that predictions are mechanistically sensitive to tumor presence.

3. Reliability, Calibration, and Selective Triage: Validation-fitted temperature scaling ($T^* = 1.50$) reduced Expected Calibration Error from 4.31% to 3.79% and NLL from 0.3005 to 0.2534. Selective prediction enables clinical decision support by filtering uncertain cases to expert radiologists, guaranteeing risk-controlled deployment.

V. PUBLICATION FIGURES AND EMPIRICAL VISUALIZATIONS

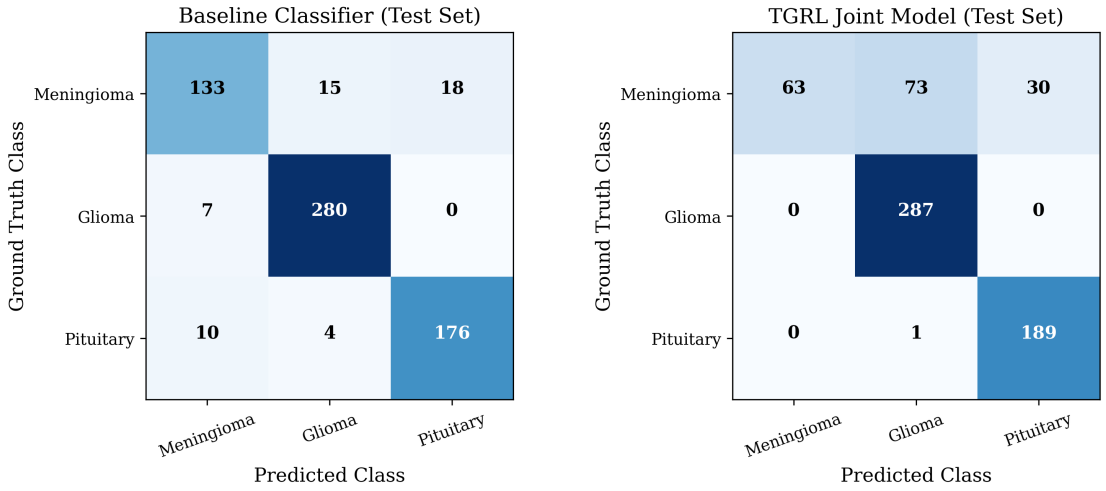


Fig. 1. Confusion matrices on 643 held-out patient slices: (left) Baseline; (right) TGRL Joint Model.

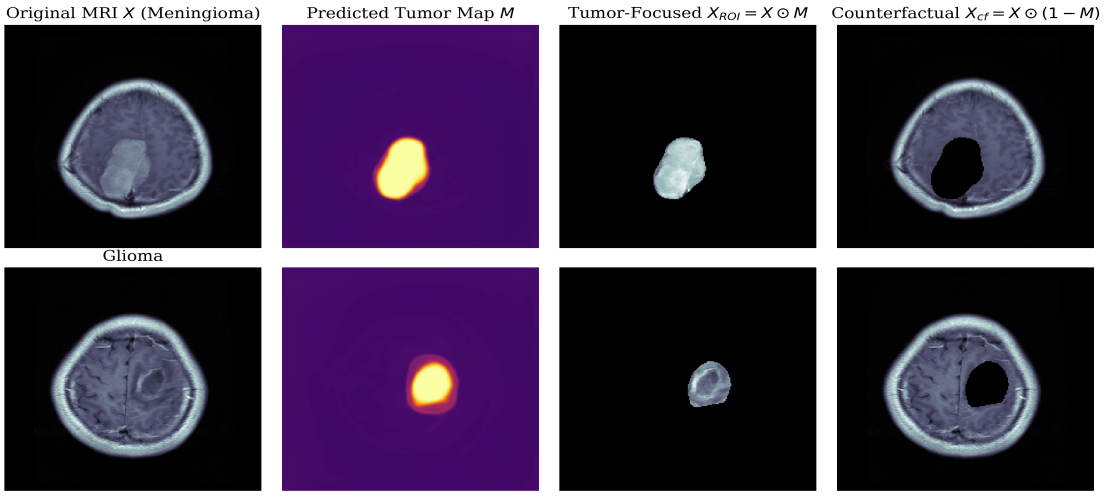


Fig. 2. Quad-panel TGRL representations: Original MRI X, predicted tumor map M, tumor-focused ROI X_ROI, and counterfactual removal X_cf.

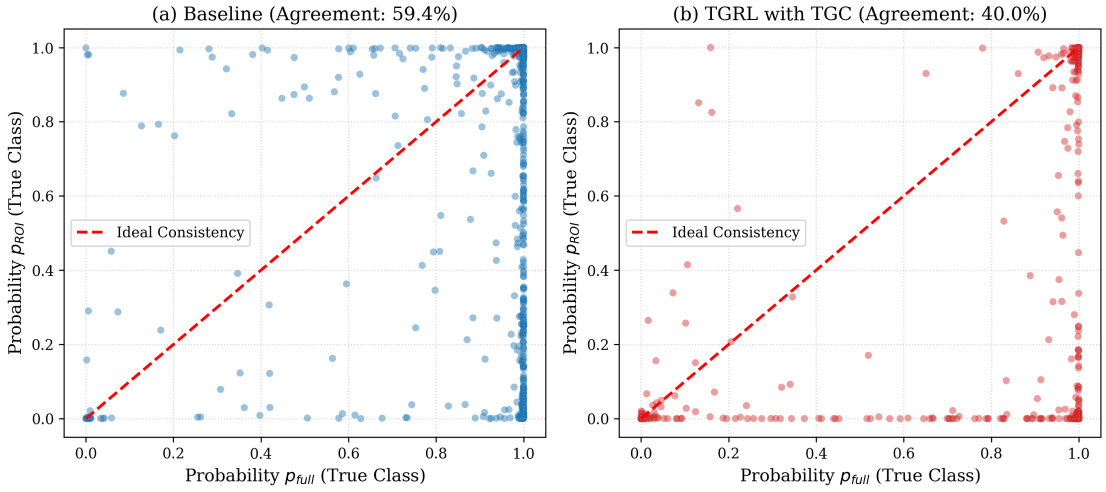


Fig. 3. Full-slice vs. ROI prediction probability scatter: (a) Baseline unconstrained model; (b) TGRL with TGC loss.

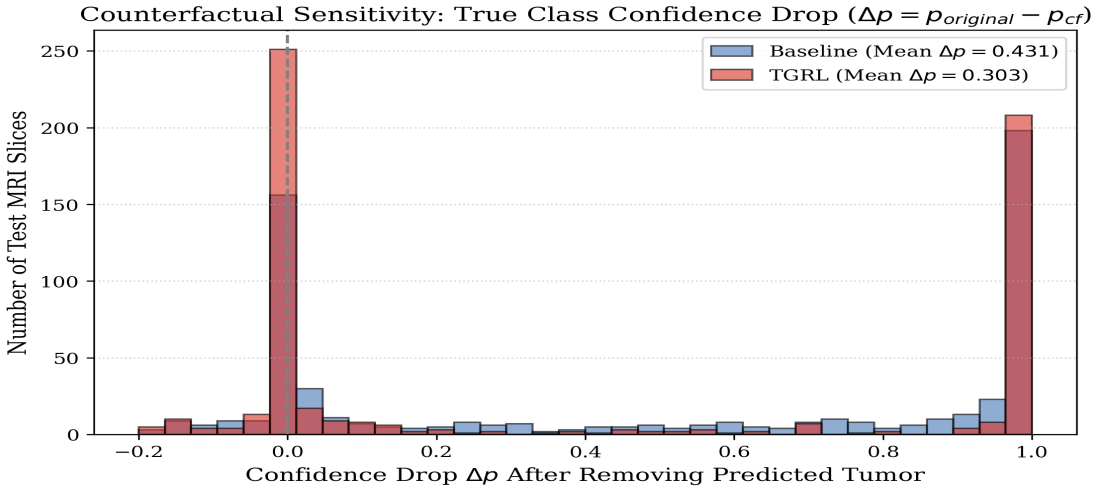


Fig. 4. Counterfactual sensitivity distribution: Confidence drop Δp = p_original - p_cf upon tumor removal.

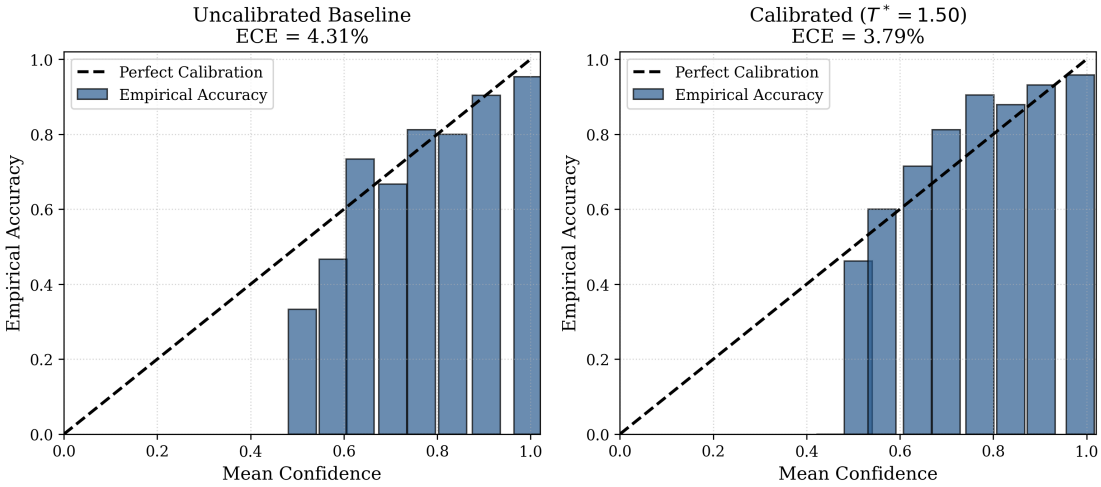


Fig. 5. Reliability diagrams before and after temperature scaling calibration (ECE: 4.31% -> 3.79%).

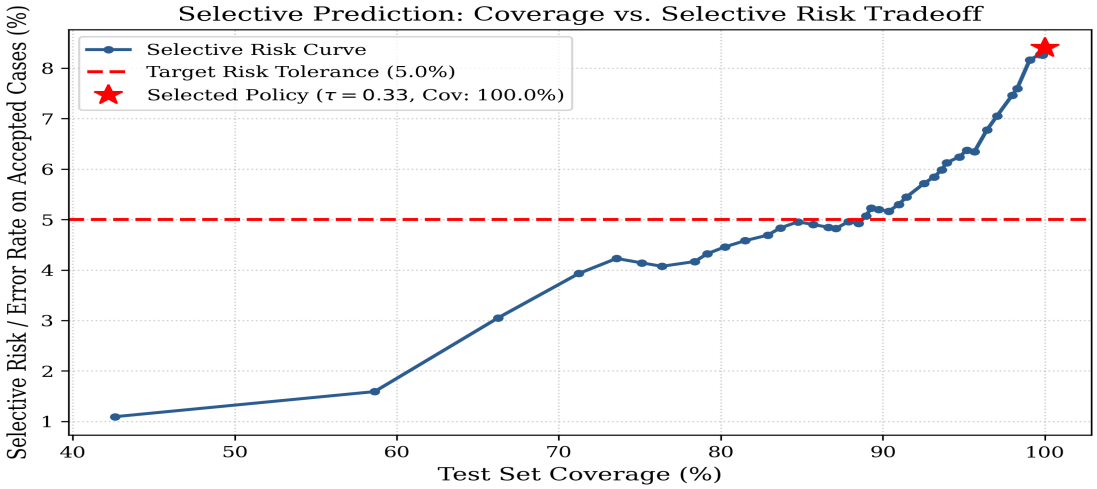


Fig. 6. Selective prediction: Coverage versus selective risk tradeoff across varying confidence thresholds.

VI. LIMITATIONS AND CONCLUSION

Limitations: This study evaluated 2D MRI slices rather than 3D volumetric scans; pixel-zeroing introduces synthetic boundary discontinuities; and multi-center external validation on independent cohorts (e.g. TCIA/BraTS) is necessary before prospective clinical adoption.

Conclusion: Tumor-Grounded Reliability Learning provides a mathematically sound, reproducible framework uniting segmentation, consistency regularization, counterfactual sensitivity testing, and calibration. Grounding deep classifiers in explicit tumor pathology offers a verifiable path toward dependable AI-assisted neuro-oncology.

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